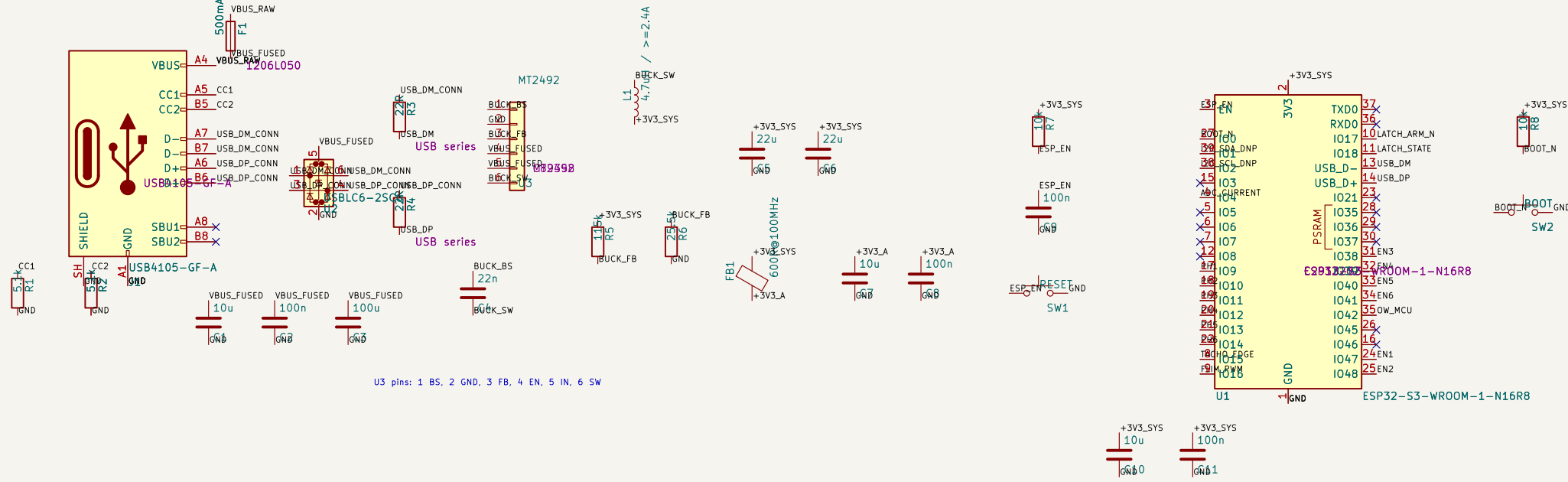


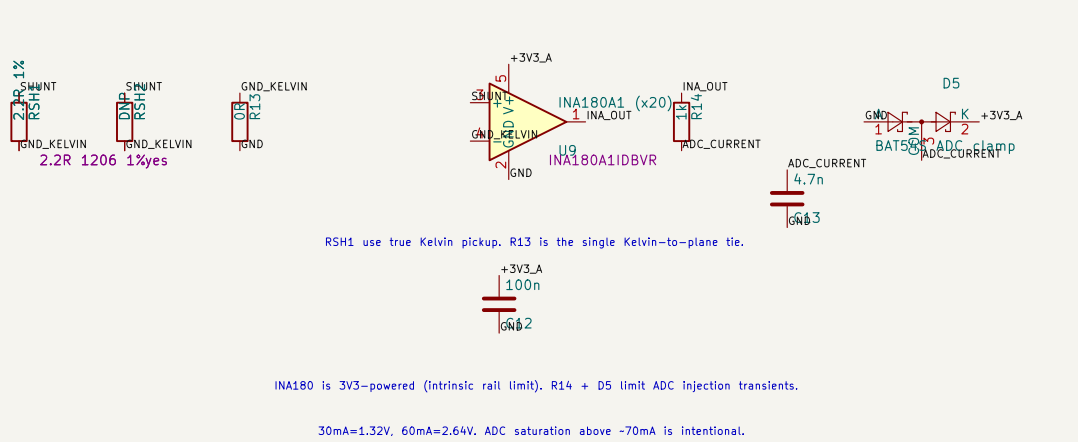
USB-C INPUT / 3V3 POWER

ESP32-S3 / USB / GPIO



U3 pins: 1 BS, 2 GND, 3 FB, 4 EN, 5 IN, 6 SW

SHARED SHUNT / CURRENT / TACHO

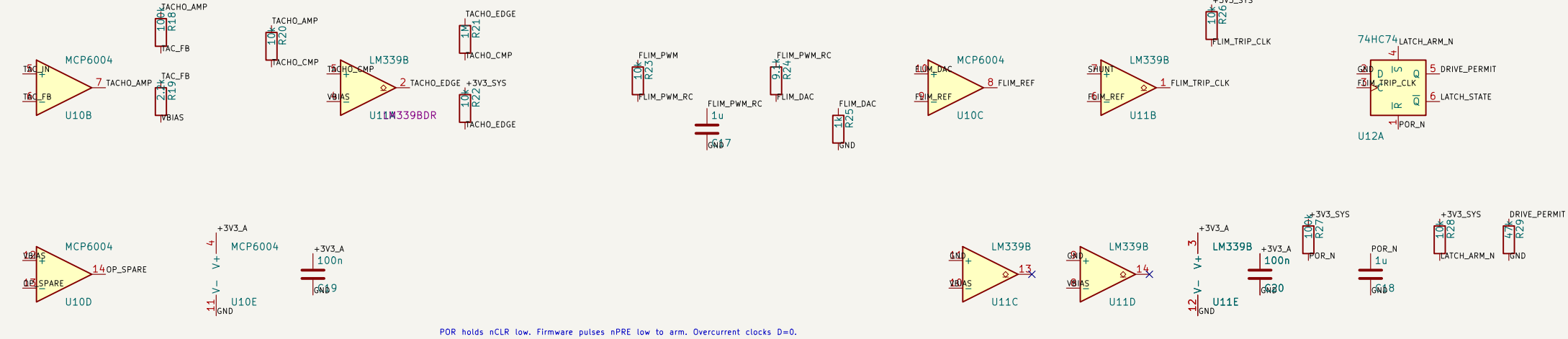


RS1H use true Kelvin pickup. R13 is the single Kelvin-to-plane tie.

INA180 is 3V3-powered (intrinsic rail limit). R14 + D5 limit ADC injection transients.

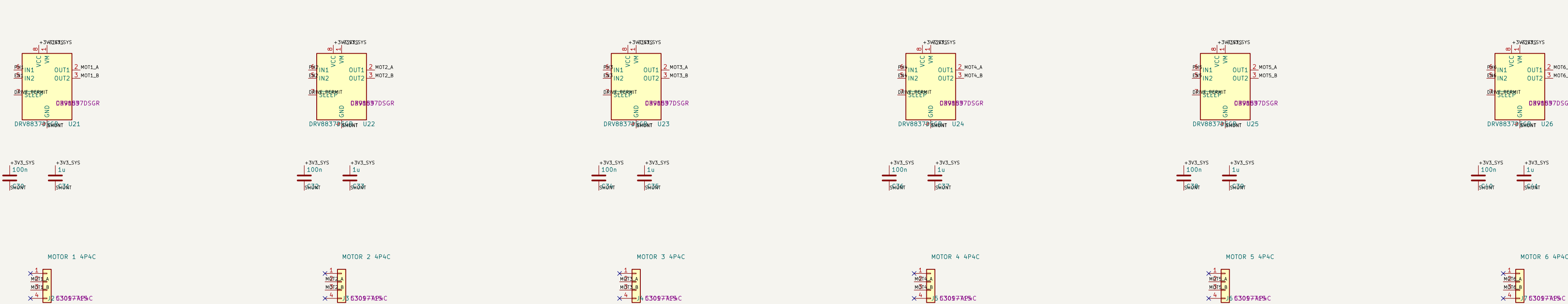
30mA=1.32V, 60mA=2.64V, ADC saturation above ~70mA is intentional.

HARDWARE FORCE LIMIT / FAIL-SAFE LATCH



PDR holds nCLR low. Firmware pulses nPRE low to arm. Overcurrent clocks D=0.

6 x DRV8837 / 4P4C MOTOR OUTPUTS



Connector pinout: 1 NC, 2 MOT_A, 3 MOT_B, 4 NC (center pair).

ONEWIRE / DNP I2C EXPANSION



OneWire pinout: 1 +3V3, 2 DQ, 3 GND, J11 pinout: GND, +3V3, SDA, SCL.